

# Rabbiya Riaz

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## Professional Summary

AI-focused Computer Science graduate with hands-on experience building **LLM-powered applications**, RAG pipelines, and AI workflow automation using LangChain, FastAPI, and vector databases. Proficient with APIs, ChromaDB, and prompt engineering for domain-specific tasks. Experienced in transformer fine-tuning, modular inference deployment, and integrating AI systems end-to-end. Currently expanding expertise in n8n workflow automation and LangGraph agent orchestration. Seeking to contribute to high-velocity AI product engineering and intelligent automation systems.

## Education

**National University of Sciences & Technology (NUST), SEECS**

BS Computer Science | Sep 2022 – Jun 2026

CGPA: 3.20/4.00

## Work Experience

### AI / ML Research Intern

School of Electrical Engineering & Computer Science (SEECS), NUST

Jul 2025 – Sep 2025

- Re-engineered a Graph Convolutional Network (**GCN**)-based action recognition system to support a new skeleton dataset (MMASD). Redesigning data preprocessing pipelines and **adapted** the **codebase** to a different skeletal data format.
- Refactored graph construction logic to correctly model **25-joint skeletal** connectivity, aligning data representation with model assumptions.
- Diagnosed and resolved **feature-model mismatches** through systematic ML debugging and data validation.
- Improved model accuracy from 10% to 80%, significantly increasing inference reliability and model usability.

## Technical Experience

### Federated Learning-Based Mental Health AI Platform (Final Year Project)

<https://github.com/rabbivari2/Federated-Learning-Based-Mental-Health-Agent>

- Fine-tuned **DistilBERT** on a domain-specific mental health dataset for multi-class intent classification using **PyTorch and Hugging Face Transformers**, achieving 65% **macro F1-score** on a highly **imbalanced multilabel** dataset.
- Designed and implemented **RAG pipelines** using ChromaDB for vector storage and semantic retrieval, experimentally comparing prompt-based inference vs fine-tuning for domain adaptation to optimize response quality.
- Conducted supervised transformer training with structured **config-based experiment logging**, evaluating performance using **F1-score**, and comparing pretrained vs fine-tuned baselines.
- Implemented **federated learning** mechanisms for privacy-preserving distributed updates, incorporating dataset cleaning, preprocessing, and anonymization aligned with responsible AI practices.
- Deployed modular **LLM inference services via FastAPI** to support scalable experimentation and prototyping.

### AI-Powered Spoken English Conversation App

- Built a real-time voice AI system integrating **Speech-to-Text**, LLaMA-based LLM inference, and Text-to-Speech pipelines into a full conversational loop, enabling context-aware dialogue generation for spoken English practice.
- Developed a cross-platform Flutter application with Firebase Auth, Firestore, and Cloud Storage for persistent session management and conversation history across devices.

### Deepfake Video Detection Using Deep Learning

- Built a spatiotemporal deepfake detection pipeline using **ResNet18** for spatial feature extraction and **Bi-LSTM** for temporal modeling, achieving improved sequence-level classification across manipulated video content.
- Engineered a **frame-level preprocessing** system that segments video into overlapping temporal sequences, resolving temporal inconsistencies in manipulated content and improving data quality for robust model training.

## Skills

**Programming:** Python, JavaScript

**LLM Systems & Advanced AI:** Large Language Models (LLMs), LangChain, LangGraph, Grok API, RAG Pipelines, Prompt Engineering, Embeddings, Inference Optimization, ChromaDB

**Machine Learning & LLM Research:** PyTorch, Hugging Face Transformers, Transformer Fine-Tuning, Supervised Learning, Hyperparameter Tuning, NLP, ML Debugging

**Backend, Deployment & Databases:** FastAPI, RESTful APIs, Docker, Git, GitHub, PostgreSQL, Render

**Data:** Dataset Preprocessing, Cleaning, Validation, Experimentation Workflows

## Certifications & Achievements

- Millennium Fellowship 2025**—Selected for a global leadership and impact program focused on social innovation.
- National PakCrypt Challenge 2025** Finalist
- 100 Days of Code:** Python Pro Bootcamp – Dr. Angela Yu